Midterm

April 22, 2025 $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} =$

1. (10 points)
For a system of linear equation, $Ax = b$, A is a full-rank 3 by 3 matrix. The solution $x = [1, 2, 3]^T$. If we put permutation matrix $P = P_{21}P_{32}$ for row/column reordering, where

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad A\begin{bmatrix} x_3 \\ x_1 \\ x_2 \end{bmatrix} = b$$

(a) What's the solution of $PAx = Pb$?
(b) What's the solution of $APx = b$?

2. (10 points) Create a 3 by 4 matrix A whose solutions to $A\mathbf{x} = 0$ are $\mathbf{s_1}$ and $\mathbf{s_2}$. The matrix A has pivot columns 1 and 3, free variables x_2 and x_4:

$$s_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} \text{ and } s_2 = \begin{bmatrix} -2 \\ 0 \\ -6 \\ 1 \end{bmatrix}.$$

Factorize A into CR, where R is the row reduced echelon form.

3. (20 pints) The equation $x - 3y - z = 0$ determines a plane in $\mathbf{R^3}$.
(a) If we formulate this equation as $Ax = 0$, What is the matrix A?
(b) What's the rank(A)?
(c) The special (null) solutions x_n are?
(d) The plane $x - 3y - z = 12$ is parallel to $x - 3y - z = 0$. One particular point x_p on this plane is $(12, 0, 0)$. All points on the plane have the form:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \ \\ 0 \\ 0 \end{bmatrix} + y\begin{bmatrix} \ \\ 1 \\ 0 \end{bmatrix} + z\begin{bmatrix} \ \\ 0 \\ 1 \end{bmatrix}.$$

Please refer to above equation, fill the complete solution to $x - 3y - z = 12$.

4. (20 points) Without computing A, find the four <u>orthogonal</u> bases for its four fundamental subspaces:

$$\underset{5\times 3}{A^T} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 3 & 0 \\ 0 & 1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\underset{3\times 5}{A} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} (= CR) = \begin{bmatrix} 1 & 3 \\ 1 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 0 & -2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

5. (20 points) Start with nine measurement b_1 to b_9, all zero, at times $t = 1, ..., 9$. The 10th measurement $b_{10} = 40$ is an outliner. Find the **best horizontal line y = C** to fit these ten points $(1,0), (2,0), ...(9,0), (10,40)$ using three options for the error E:
(a) Least squares $E_2 = e_1^2 + ... + e_{10}^2$.
(b) Least maximum error $E_\infty = |e_{max}|$.
(c) Least sum of errors $E_1 = |e_1| + ... + |e_{10}|$.
(d) What's the physical significance of the Least Squares (L2-norm), Least maximum error (Lmax-norm), and Least sum of errors (L1-norm), why we design three different merits for the error?

6. (20 points)
6 8 4
(a) Project the vector $b^T = (3, 4, 4)$ onto the line through $a = (2, 2, 1)$ and then onto the plane that also contains $a^* = (1, 0, 0)$.
(b) check the first error vector $e = b - p$ is perpendicular to a, and the second error $e^* = b - p^*$ is also perpendicular to a^*.
(c) Find a 3 by 3 projection matrix P onto the plane containing a and a^*
(d) Find the complementary projection matrix $P_\perp$, which project the vector $b^T = (3, 4, 4)$ onto zero vector.
0 1 -1

7. **(BONUS)** Explain the Linear transformation Ax and Inverse (or Pseudo-inverse) of $Ax = b$ in big picture of the four subspaces of A. [Hint: you can use an over-determined case ($m > n = r$) such as Problem 6].

$$AA^+x = p$$

$$p = Pb$$
$$= AA^+b$$

$$\begin{array}{c} \\ m \end{array}\overset{n}{\begin{bmatrix} & \end{bmatrix}} \begin{bmatrix} \\ \end{bmatrix}_n = \overset{m}{\begin{bmatrix} \\ \end{bmatrix}}$$